Longest Arithmetic Subsequence (medium)

Problem Statement

Given an array `nums` containing positive integers, return the maximum length of a subsequence that forms an `arithmetic progression`.

- A `subsequence` is an array that can be formed from `nums` by deleting 0 or more elements without changing the order of the remaining elements.
- An arithmetic progression (AP) is a sequence of numbers in which the difference between any two consecutive elements is constant. In short, `seq[i + 1] - seq[i]` should be same for all `0 <= i < seq.length - 2`.

Examples

- **Example 1**:
  - **Input**: `[8, 12, 6, 4, 2]`
  - **Expected Output**: `4`
  - **Justification**: The subsequence `[8, 6, 4, 2]` forms the longest arithmetic subsequence with a common difference of **-2**, thus the longest arithmetic subsequence has a length **4.**
- **Example 2**:
  - **Input**: `[5, 10, 15, 7, 11, 5]`
  - ****Expected Output: `3`
  - Justification:** The subsequence **[5, 10, 15]** forms the longest arithmetic subsequence with a common difference of **5.**
- **Example 3**:
  - **Input**: `[1, 7, 10, 15, 27, 29]`
  - **Expected Output**: `3`
  - **Justification**: The subsequence **[1, 15, 29]** forms the longest arithmetic subsequence with a common difference of **14.**

Try it yourself

Try solving this question here:

Java

```java
import java.util.HashMap;
import java.util.Map;

public class Solution {
    // Method to find the length of the longest arithmetic subsequence
    public int longestArithSeqLength(int[] A) {
        // ToDo: Write Your Code Here.
        return 0;
    }
}
```

Test Case | Run Code Result | Stdout

[8, 12, 6, 4, 2]

Use Example Testcases | Run | Submit

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Solution: Longest Arithmetic Subsequence

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